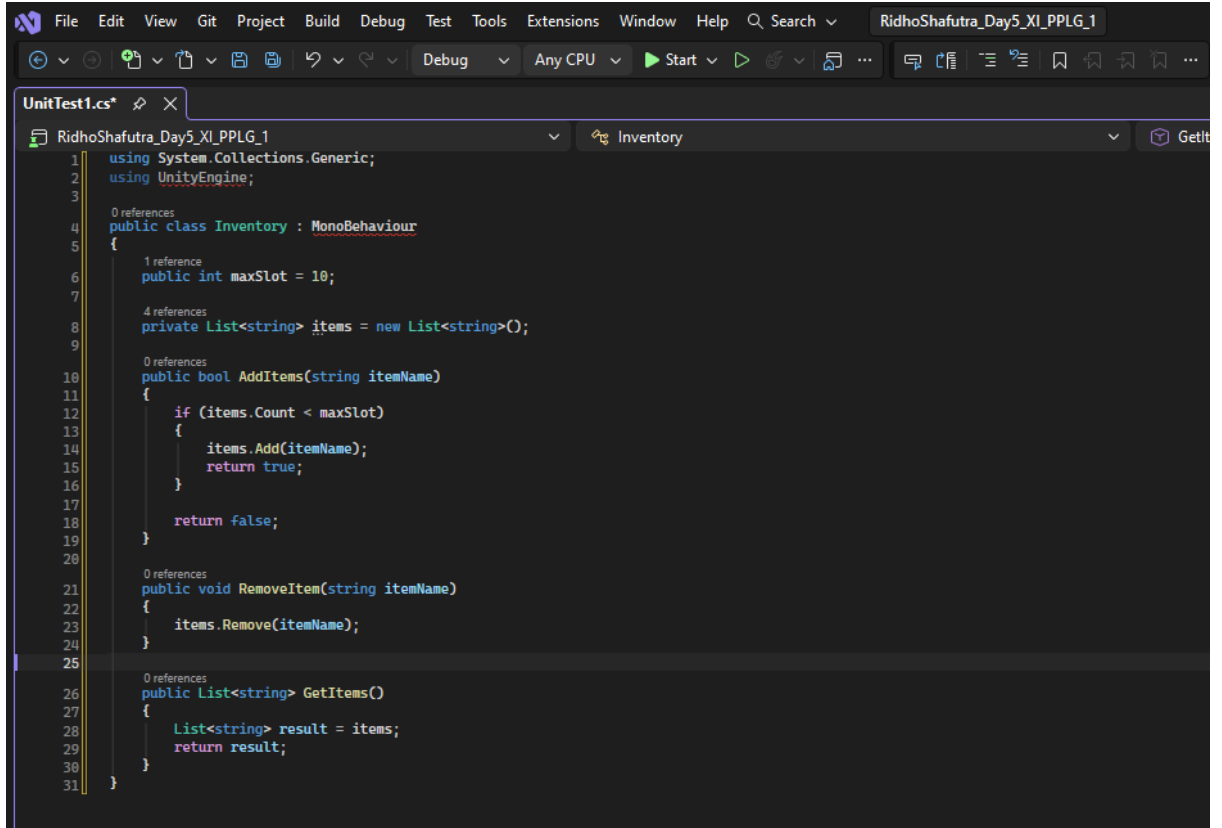


TUGAS BLOKING DAY 5 WEEK 2

Tantangan 1-8

RIDHO SHAFUTRA

1. Tantangan 1



The screenshot shows the Visual Studio IDE with a C# file named 'UnitTest1.cs'. The code defines an 'Inventory' class that inherits from 'MonoBehaviour'. It includes a 'maxSlot' property, a private 'items' list, and three methods: 'AddItems', 'RemoveItem', and 'GetItems'. The 'AddItems' method checks if the current count is less than 'maxSlot' before adding an item. The 'RemoveItem' method removes an item from the list. The 'GetItems' method returns the current list of items.

```
1 using System.Collections.Generic;
2 using UnityEngine;
3
4 0 references
5 public class Inventory : MonoBehaviour
6 {
7     1 reference
8     public int maxSlot = 10;
9
10    4 references
11    private List<string> items = new List<string>();
12
13    0 references
14    public bool AddItems(string itemName)
15    {
16        if (items.Count < maxSlot)
17        {
18            items.Add(itemName);
19            return true;
20        }
21        return false;
22    }
23
24    0 references
25    public void RemoveItem(string itemName)
26    {
27        items.Remove(itemName);
28    }
29
30    0 references
31    public List<string> GetItems()
32    {
33        List<string> result = items;
34        return result;
35    }
36 }
```

2. Tantangan 2

```

31 public enum Element
32 {
33     2 references
34     Fire,
35     2 references
36     Earth,
37     2 references
38     Water
39 }
40
41 0 references
42 public class DamageSystem : MonoBehaviour
43 {
44     0 references
45     public float CalculateDamage(
46         float baseDamage,
47         Element attackerElement,
48         Element defenderElement,
49         float defense,
50         bool isCritical)
51     {
52         // 1. Damage dasar
53         float damage = baseDamage;
54
55         // 2. Hitung efektivitas elemen
56         float elementalMultiplier = 1f;
57
58         if (attackerElement == defenderElement)
59         {
60             // Tipe sama = 0.5x
61             elementalMultiplier = 0.5f;
62         }
63         else if (
64             (attackerElement == Element.Fire && defenderElement == Element.Earth) ||
65             (attackerElement == Element.Earth && defenderElement == Element.Water) ||
66             (attackerElement == Element.Water && defenderElement == Element.Fire)
67         )
68         {
69             // Elemen unggul = 1.5x
70             elementalMultiplier = 1.5f;
71         }
72
73         damage *= elementalMultiplier;
74
75         // 3. Critical hit = 2x
76     }
77 }

```

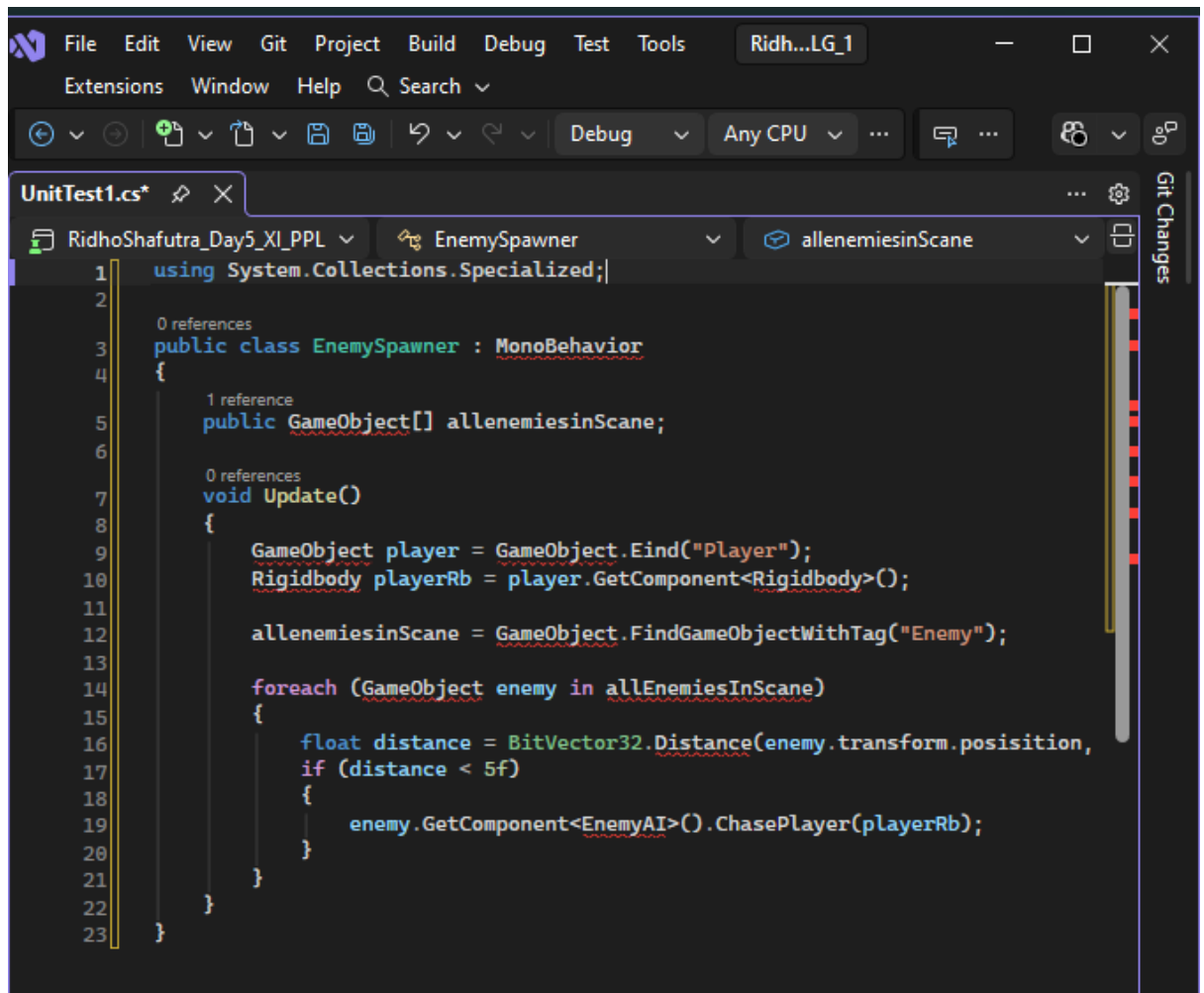
3. Tantangan 3

```

4 1 reference
5 public abstract class GameCharacter : MonoBehaviour
6 {
7     3 references
8     public string characterName;
9     2 references
10    public int maxHP = 100;
11    5 references
12    public int currentHP = 100;
13
14    // Setiap karakter wajib memiliki Attack sendiri
15    1 reference
16    public abstract void Attack();
17
18    // Method menerima damage
19    0 references
20    public virtual void TakeDamage(int damage)
21    {
22        currentHP -= damage;
23
24        if (currentHP < 0)
25        {
26            currentHP = 0;
27        }
28
29        Debug.Log(characterName + " menerima " + damage + " damage.");
30    }
31
32    // Mengecek apakah karakter masih hidup
33    0 references
34    public bool IsAlive()
35    {
36        return currentHP > 0;
37    }
38 }
39
40 // ===== PLAYER =====
41
42 0 references
43 public class Player : GameCharacter
44 {
45     2 references
46     public int level = 1;
47     3 references
48     public int experience = 0;
49
50     0 references
51     private void Start()
52     {
53         characterName = "Player";
54         maxHP = 100;
55         currentHP = maxHP;
56     }
57 }

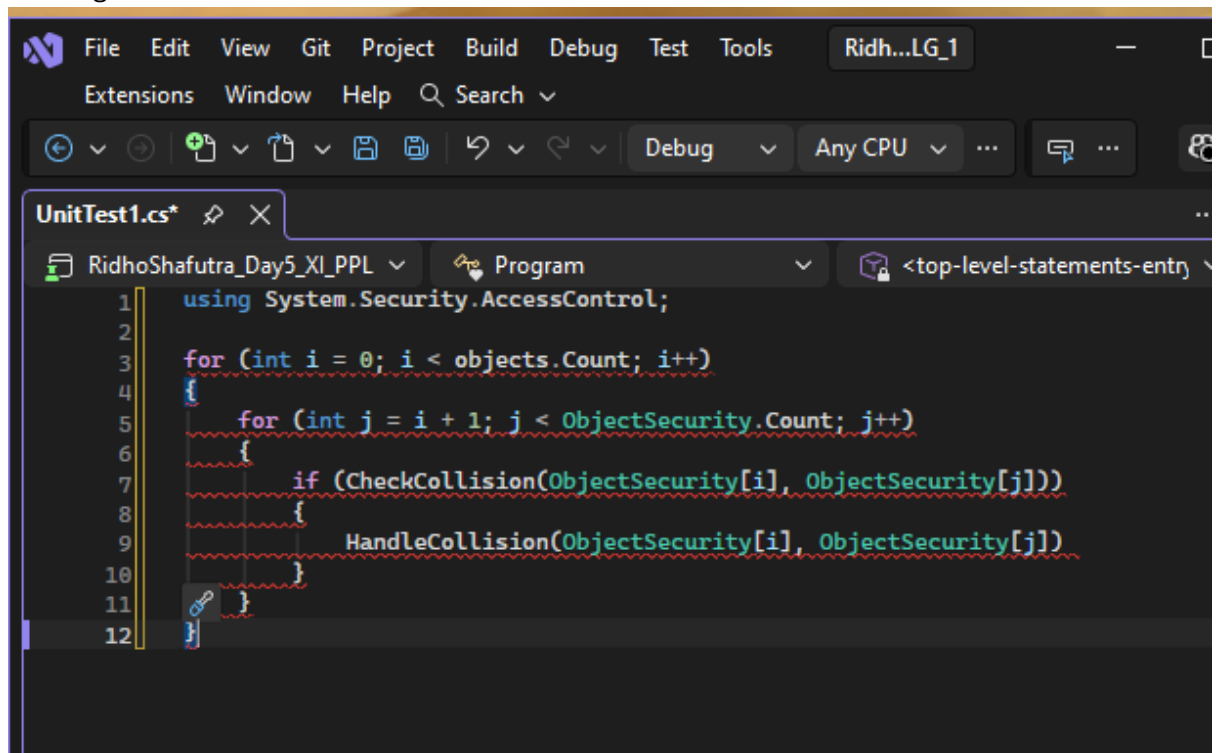
```

4. Tantangan 4



```
1 using System.Collections.Specialized;
2
3 public class EnemySpawner : MonoBehaviour
4 {
5     public GameObject[] allEnemiesInScene;
6
7     void Update()
8     {
9         GameObject player = GameObject.Find("Player");
10        Rigidbody playerRb = player.GetComponent<Rigidbody>();
11
12        allEnemiesInScene = GameObject.FindGameObjectsWithTag("Enemy");
13
14        foreach (GameObject enemy in allEnemiesInScene)
15        {
16            float distance = BitVector32.Distance(enemy.transform.position,
17            if (distance < 5f)
18            {
19                enemy.GetComponent<EnemyAI>().ChasePlayer(playerRb);
20            }
21        }
22    }
23 }
```

5. Tantangan 5



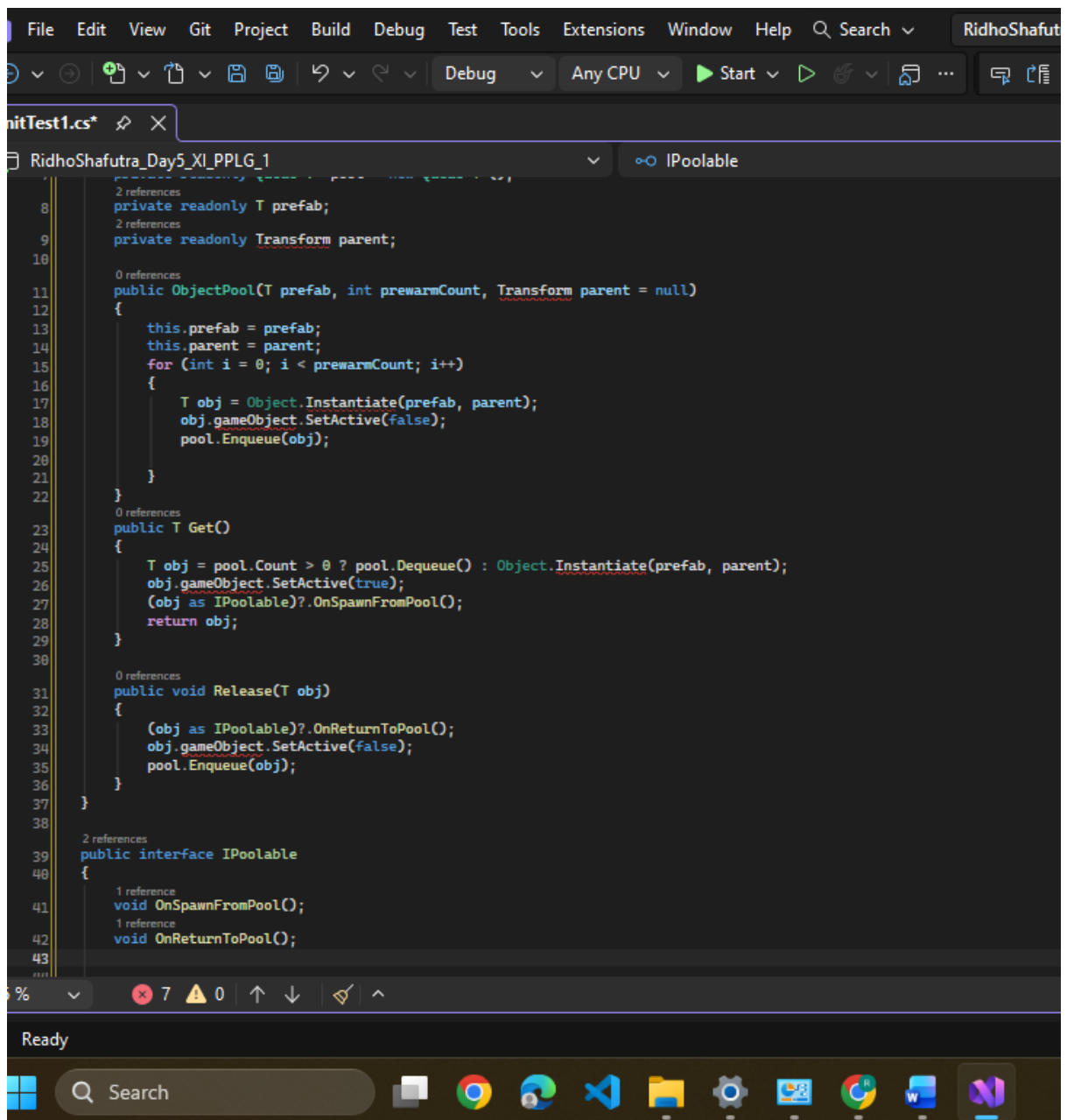
```
1 using System.Security.AccessControl;
2
3 for (int i = 0; i < objects.Count; i++)
4 {
5     for (int j = i + 1; j < ObjectSecurity.Count; j++)
6     {
7         if (CheckCollision(ObjectSecurity[i], ObjectSecurity[j]))
8         {
9             HandleCollision(ObjectSecurity[i], ObjectSecurity[j])
10        }
11    }
12 }
```

6. Tantangan 6

```
RidhoShafutra_Day5_XI_PPL  SaveSystem  AddGold(int amou

1  2 references
   public class PlayerData
2  {
3      2 references
       public int gold;
4      1 reference
       public int level;
5  }
6
7  0 references
   public class SaveSystem : MonoBehaviour
8  {
9      1 reference
       private PlayerData data = new PlayerData { gold = 0, level = 1 };
10
11     0 references
       public void AddGold(int amount)
12     {
13         data.gold += amount;
14     }
15 }
```

7. Tantangan 7



8. Tantangan 8

File Edit View Git Project Build Debug Test Tools Extensions Window Help Search RidhoShafutra_Day5_XI.P

UnitTest1.cs*

RidhoShafutra_Day5_XI_PPLG_1 AttackSystem

```
4 // ===== GSP INHERITANCE =====
5
6 public abstract class GameCharacter : MonoBehaviour
7 {
8     public string characterName;
9     public int hp = 100;
10
11     public abstract void Attack();
12
13     0 references
14     public virtual void TakeDamage(int damage)
15     {
16         hp -= damage;
17         if (hp < 0)
18             hp = 0;
19         Debug.Log(characterName + " terkena " + damage + " damage.");
20     }
21
22     public bool IsAlive()
23     {
24         return hp > 0;
25     }
26 }
27
28 // ===== PLAYER =====
29
30 public class Player : GameCharacter
31 {
32     public int level = 1;
33     public int experience = 0;
34
35     private void Start()
36     {
37         characterName = "Player";
38     }
39
40     public override void Attack()
41     {
42         Debug.Log(characterName + " menyerang musuh!");
43     }
44
45     public void AddExperience(int exp)
46     {
47         experience += exp;
48         if (experience >= 100)
49         {
50             level++;
51             experience = 0;
52             Debug.Log("Player naik ke level " + level);
53         }
54     }
55 }
56
57
```

51 % 13 0

Ready